

PIYUSH SHAH

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SUMMARY

Instrumentation and Control Engineering Technology student graduating December 2026 from Red River College, with hands-on co-op experience in building automation and controls, PLC programming on Allen-Bradley CompactLogix, and applied process instrumentation. Background in Computer Science adds data analysis, Python scripting, and machine-learning capabilities uncommon in instrumentation candidates. Seeking a full-time role in industrial instrumentation or process automation where strong technical fundamentals and a practical project portfolio translate directly to plant-floor contribution.

EDUCATION

Instrumentation & Control Engineering Technology (Diploma)

Sep 2024 – Dec 2026

Red River College | Winnipeg, MB

- Coursework: process measurement, control systems, PLC programming, industrial instrumentation, power distribution, schematic design, and process control loops
- Extensive lab work with 4–20 mA signal loops, sensor calibration, PID/PIDE tuning, Allen-Bradley CompactLogix, FactoryTalk View Studio SE HMI integration, and ISA-18.2 alarm implementation

Bachelor of Computer Science (Coursework Completed)

Sep 2022 – Apr 2024

University of Manitoba | Winnipeg, MB

- Completed two years of coursework in software development, algorithms, data structures, and applied mathematics
- Developed strong problem-solving and data analysis skills in C++, Python, and Java

TECHNICAL SKILLS

PLCs & Control Platforms: Allen-Bradley CompactLogix, Studio 5000 Logix Designer, FactoryTalk View Studio SE, Arduino

Instrumentation: 4–20 mA signal loops, RTD/thermocouple/pressure/level transmitters, sensor calibration, loop commissioning, P&ID interpretation, ISA-5.1 symbology

Control & Protocols: PID/PIDE tuning (IMC, Ziegler-Nichols), BACnet MS/TP, Modbus RTU/TCP, OPC-UA, RS-485

Electrical & Wiring: Control panel assembly, schematic interpretation, wire routing/termination, soldering, connector assembly, functional testing

Programming & Software: Python (NumPy, scikit-learn, Pandas, Matplotlib), C++, MATLAB, AutoCAD, Visio, Multisim, Solid Edge, Excel

Data & Monitoring: Real-time data acquisition, vibration/current/temperature monitoring, Grafana, InfluxDB, SCADA dashboards

Documentation: Loop diagrams, P&IDs, instrument datasheets, test reports, technical manuals

WORK EXPERIENCE

BAS Controls Specialist (Co-op)

May 2026 – Present

ICS — Integrated Control Solutions | Winnipeg, MB

- Commissioning and programming building automation systems using BACnet MS/TP and Modbus RTU fieldbus protocols on RS-485 networks
- Configuring and troubleshooting field devices including HVAC sensors, actuators, and controllers across commercial building installations
- Interpreting control system drawings and sequences of operation to implement and verify correct system behaviour on-site
- Collaborating with project engineers to resolve integration issues between BAS controllers and third-party equipment

Electrical Technician

Jun 2025 – Sep 2025

Tamarack Industries | Winnipeg, MB

- Assembled and wired 50+ industrial control panels and chassis harnesses for glycol heating equipment, interpreting schematic drawings to ensure compliance with client specifications and industrial standards
- Performed wire cutting, routing, crimping, and connector termination; conducted functional testing and inspections, achieving near-zero defects in completed units
- Collaborated with engineers to troubleshoot production issues, minimizing downtime and improving assembly efficiency

TECHNICAL PROJECTS

Cascaded PIDE Level Control System with FactoryTalk HMI

Winter 2026

Allen-Bradley CompactLogix | Studio 5000 | FactoryTalk View Studio SE

- Designed and implemented a cascaded PID control system in Allen-Bradley CompactLogix to regulate water tank level under variable inflow disturbances — outer loop controlling tank level, inner loop controlling inlet flow rate
- Configured PIDE instruction with engineering-unit scaling, anti-windup, output limiting, and bumpless auto/manual transfer; tuned both loops using CompactLogix built-in autotune and validated against step-response targets
- Built a FactoryTalk View Studio SE operator HMI featuring animated tank level display, live PV/SP/output trends, setpoint entry, start/stop controls, and ISA-18.2-style tiered alarm banner

AI-Assisted Predictive Maintenance System

Dec 2025 – Feb 2026

Python | scikit-learn | Arduino | ADXL345 | ACS712 | MLX90614

- Designed a motor health monitoring system integrating an ADXL345 accelerometer, MLX90614 infrared temperature sensor, and ACS712 current sensor with Arduino for real-time multi-parameter data acquisition
- Extracted time-domain and frequency-domain features from vibration and current signals; trained a Random Forest classifier to distinguish healthy from degraded motor states, achieving measurable separation between fault classes
- Implemented tiered alarm logic (advisory/warning/critical/emergency shutdown) consistent with ISA-18.2 alarm management principles; visualized live sensor data and fault indicators on a Python Dash SCADA-style dashboard

RTD Temperature Transmitter — 4–20 mA Signal Conditioning Circuit

Oct – Nov 2025

Analog Circuit Design | Instrumentation Amplifier | Current Loop

- Designed and built an RTD-based temperature transmitter converting small resistance changes to a standard 4–20 mA industrial signal, using an instrumentation amplifier to accurately amplify the low-level differential RTD signal
- Implemented transistor-based current loop driver with 250 Ω load resistor; verified correct loop current across full temperature range with multi-point measurement and calibration checks

Sequential Function Chart — Automated Gate Control System

Winter 2026

Allen-Bradley CompactLogix | Studio 5000 | IEC 61131-3 SFC

- Programmed an IEC 61131-3 Sequential Function Chart in Allen-Bradley CompactLogix implementing a multi-state gate control sequence: normal operation, ticket issuance, barrier actuation, alarm activation, and maintenance/fault mode
- Designed safe-state transition logic ensuring deterministic fault handling and alarm annunciation on sensor fault or sequence timeout; implemented maintenance mode with system status indication

Closed-Loop PID Temperature Control — IMC Tuning

Jan 2026

Arduino | PWM | MOSFET | IMC Tuning | Excel Trending

- Implemented a closed-loop PID temperature controller using a PWM-driven MOSFET to linearly vary power to a resistive heater; performed open-loop step test to identify FOPDT process model parameters

- Applied Internal Model Control (IMC) tuning methodology to derive P, I, D parameters from measured process gain, time constant, and dead time; validated closed-loop step response and documented tuning in Excel trend charts

PROFESSIONAL ATTRIBUTES

- Strong foundation in both classical instrumentation (4–20 mA, PID, P&ID) and modern data/programming skills (Python, ML, OPC-UA) — a combination uncommon in instrumentation candidates
- Hands-on co-op experience commissioning real BAS systems with live fieldbuses; comfortable working in technical field environments
- Detail-oriented with systematic troubleshooting approach developed across panel wiring, sensor calibration, and control loop commissioning work
- Effective communicator and team contributor; proven ability to collaborate with engineers and tradespeople on-site